

Software Requirements Specification (SRS)

Online Food Delivery & Table Reservation System

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(IEEE 830 Style)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the “Online Food Delivery and Table Reservation System”. It serves as a reference for stakeholders, developers, testers, and project managers to understand system behavior, constraints, and expectations.

1.2 Document Conventions

- IEEE-style section numbering is used
- Functional requirements are labeled as FR-#
- Non-functional requirements are labeled as NFR-#
- The word shall indicate a mandatory requirement

Intended Audience and Reading Suggestions

This document is intended for:

- Project stakeholders and clients
- Software developers and designers
- Project evaluators and faculties
- Deployment and maintenance teams

Readers may focus on sections relevant to their role.

1.4 Project Scope

The system is a Online food delivery and table management system that allows customers to browse menus, order food, make online payments, and reserve restaurant tables. Administrators can manage menus, view and process orders, and update order statuses.

1.5 References

- IEEE Std 830-1998 – Software Requirements Specification
 - IEEE Std 1016-2009 – Software Design Description
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2. Overall Description

2.1 Product Perspective

The application is a web-based and mobile-responsive system consisting of:

- Customer interface
- Admin dashboard
- Backend services

- Payment gateway integration

2.2 Product Functions

Major functions include:

- User registration and login
- Menu browsing and food ordering
- Online payment processing
- Table reservation management
- Order status tracking
- Admin order and menu management

2.3 User Classes and Characteristics

- Customer – Browses menu, places orders, makes payments, and reserves tables
- Admin – Manages menu, views orders, updates order status, and manages reservations

2.4 Operating Environment

- Web browser (Chrome, Edge, Firefox)
- Mobile-responsive UI
- Backend server with relational database

2.5 Design and Implementation Constraints

- Secure payment gateway integration
- HTTPS communication required
- Role-based access control

2.6 User Documentation

- User guide for customers
- Admin dashboard manual

2.7 Assumptions and Dependencies

- Internet connectivity is available
- Payment gateway services are operational

3. System Features (Functional Requirements)

FR-1 User Registration and Login

- The system shall allow users to register using email or phone number.
- The system shall allow users to log in securely.

FR-2 Menu Browsing

- The system shall display food categories and menu items.
- The system shall show price, description, and availability of items.

FR-3 Food Ordering

- The system shall allow users to add items to a cart.
- The system shall allow users to place food orders.

FR-4 Payment Processing

- The system shall support online payment methods.
- The system shall confirm payment status after transaction.

FR-5 Order Tracking

- The system shall allow users to track order status.
- The system shall notify users when order status changes.

FR-6 Table Reservation

- The system shall allow users to reserve tables by date and time.
- The system shall confirm or reject reservations.

FR-7 Admin Order Management

- The admin shall view all customer orders.
- The admin shall update order status (Preparing, dispatched, Delivered).

FR-8 Admin Menu Management

- The admin shall add, update, or remove menu items.

4. External Interface Requirements

4.1 User Interfaces

- Customer web interface
- Admin dashboard

4.2 Hardware Interfaces

- Mobile phones, tablets, laptops, desktops

4.3 Software Interfaces

- Payment gateway API
- Database management system

4.4 Communication Interfaces

- Secure HTTP (HTTPS)

5. Non-Functional Requirements

NFR-1 Performance

- The system shall load pages within 3 seconds under normal load.

NFR-2 Security

- User data and payments shall be encrypted.

NFR-3 Reliability

- The system shall maintain consistent order processing.

NFR-4 Usability

- The system shall have a user-friendly interface.

NFR-5 Scalability

- The system shall support increasing number of users.

NFR-6 Maintainability

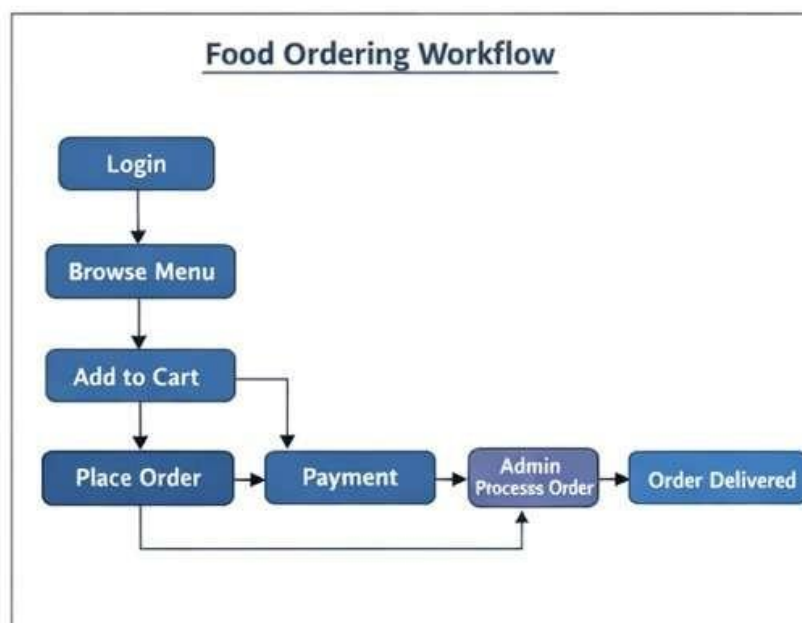
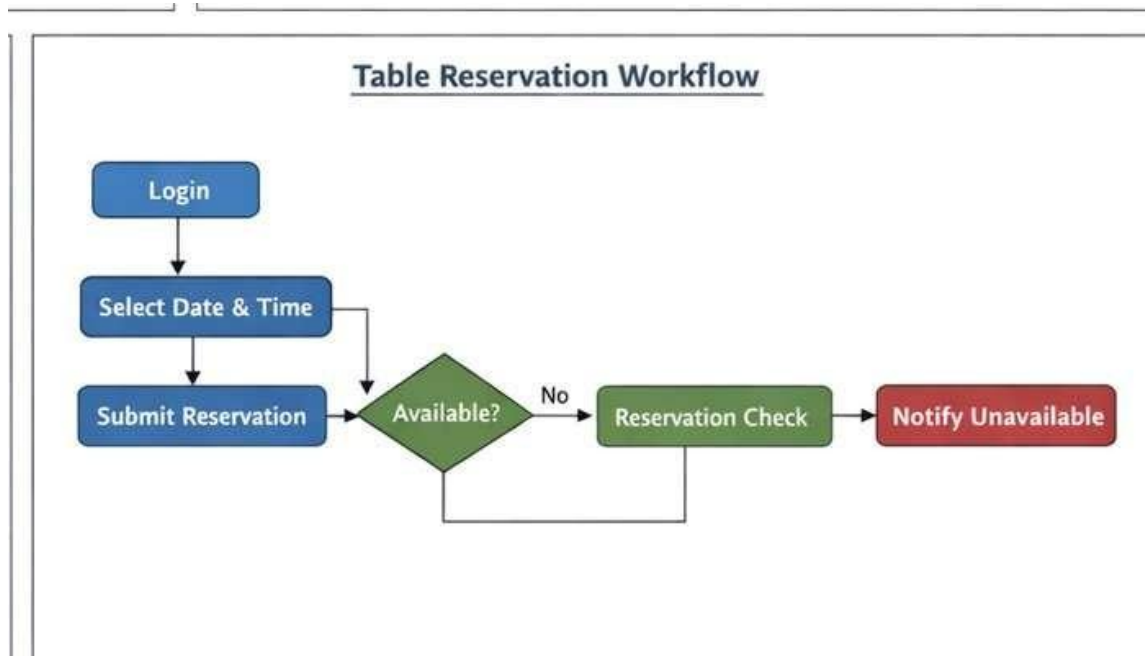
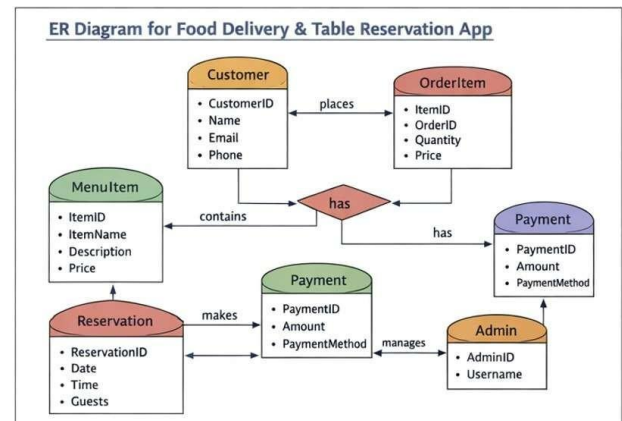
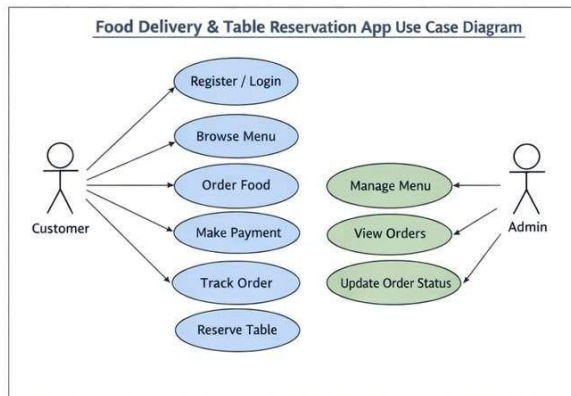
- The system shall follow modular and clean code practices.
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6. System Architecture Overview

The system follows a layered architecture:

- Presentation Layer (UI)
 - Application Layer (Business Logic)
 - Data Layer (Database)
 - Integration Layer (Payment & Notifications)
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7. System Models



8. Validation and Acceptance Criteria

- All functional requirements must pass test cases
 - Successful payment and order flow verification
 - Admin order management validation
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9. Appendices

Appendix A – Glossary

- Admin – Restaurant management user
- Customer – End user ordering food

Appendix B – Sample Data

- Sample menu items and orders

Appendix C – Future Enhancements

- Mobile app version
- Loyalty and discount system